

No.	Solution	Remarks
1(a)	AB: The ball moves down the slope with constant acceleration. BC: The ball decelerates uniformly to rest before accelerating uniformly in the opposite direction, up the slope. CD: The ball moves up the slope with constant deceleration before coming to a stop at D.	[1] [1] [1]
1(b)(i)	$a = \frac{v - u}{t} = \frac{3.8 - (-3.8)}{0.2} = 38 \text{ m s}^{-2}$	[1] correct substitution [1] answer
1(b)(ii)	length of incline = area under the graph AB = 0.5 (0.9)(3.8) = 1.7 m	[1] answer
2(a)	As the car speeds up, the resistive forces acting on it will increase	[1] drag